

Mini Exercise 1: One-Hot Coding

1. Load “bike_types_times.csv” into your Python program. The first few lines of the file are shown below:

```
Price,Time_100m_s,Road_Bike_Type
9874.55,8.62,Aero
6230.91,11.95,Endurance
10222.88,10.01,Climber
10765.24,10.15,Climber
6789.45,8.93,Aero
3900.56,11.67,Endurance
11988.99,9.54,Aero
9485.11,10.47,Climber
3345.78,12.12,Endurance
7240.90,9.33,Aero
8922.44,11.25,Endurance
6000.30,9.80,Climber
8562.50,9.92,Climber
11200.45,9.01,Aero
7180.90,9.46,Aero
3001.45,12.49,Endurance
8912.15,8.97,Aero
9223.62,10.63,Climber
```

2. The features are the Price of a bike, its 100m sprint time and its type – Aero, Climber or Endurance.
3. One-Hot Encode the last feature, Road_Bike_Type, such that your Pandas DataFrame looks like this when printed:

One-hot encoded dataframe:

	Price	Time_100m_s	Road_Bike_Type_Aero	Road_Bike_Type_Climber	Road_Bike_Type_Endurance
0	9874.55	8.62	1.0	0.0	0.0
1	6230.91	11.95	0.0	0.0	1.0
2	10222.88	10.01	0.0	1.0	0.0
3	10765.24	10.15	0.0	1.0	0.0
4	6789.45	8.93	1.0	0.0	0.0